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14. Why is there no lunar eclipse and solar eclipse every month?
15. How will you prove that Earth itself is spinning?

III. Long Answer Questions

1. Discuss the important features of the law of gravitation.
2. Explain how Newton arrived at his law of gravitation from Kepler's third law.
3. Explain how Newton verified his law of gravitation.
4. Derive the expression for gravitational potential energy.
5. Prove that at points near the surface of the Earth, the gravitational potential energy of the object is $U = mgh$
6. Explain in detail the idea of weightlessness using lift as an example.
7. Derive an expression for escape speed.
8. Explain the variation of g with lattitude.
9. Explain the variation of g with altitude.
10. Explain the variation of g with depth from the Earth's surface.
11. Derive the time period of satellite orbiting the Earth.
12. Derive an expression for energy of satellite.
13. Explain in detail the geostationary and polar satellites.
14. Explain how geocentric theory is replaced by heliocentric theory using the idea of retrograde motion of planets.
15. Explain in detail the Eratosthenes method of finding the radius of Earth.
16. Describe the measurement of Earth's shadow (umbra) radius during total lunar eclipse

IV. Conceptual Questions

1. In the following, what are the quantities which that are conserved?
   a) Linear momentum of planet
   b) Angular momentum of planet
   c) Total energy of planet
   d) Potential energy of a planet
2. The work done by Sun on Earth in one year will be
   a) Zero b) Non zero
   c) positive d) negative
3. The work done by Sun on Earth at any finite interval of time is
   a) positive, negative or zero
   b) Strictly positive
   c) Strictly negative
   d) It is always zero
4. If a comet suddenly hits the Moon and imparts energy which is more than the total energy of the Moon, what will happen?
5. If the Earth's pull on the Moon suddenly disappears, what will happen to the Moon?
6. If the Earth has no tilt, what happens to the seasons of the Earth?
7. A student was asked a question 'why are there summer and winter for us? He replied as 'since Earth is orbiting in an elliptical orbit, when the Earth is very far away from the Sun(aphelion) there will be winter, when the Earth is nearer to the Sun(perihelion) there will be winter'. Is this answer correct? If not,